

Devadarshini Pazhanivel Thenmozhi

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Education

Northeastern University

Master of Science in Artificial Intelligence

Sep 2024 – Dec 2026

Boston, Massachusetts

Anna University

Bachelor of Technology in Artificial Intelligence and Data Science

Sep 2020 – May 2024

Chennai, India

Work Experience

ProjectPathAI

Artificial Intelligence Intern

Jan 2026 –Present

Boston, Massachusetts

- Designed and implemented a multi-stage **Corrective RAG (CRAG)** pipeline using **LangGraph**, implementing LLM-based relevance grading and ambiguity-aware query refinement improving answer relevance by **20%** over baseline RAG in offline evaluation.
- Built distributed AI workflows with **Celery workers** and **Redis-backed queues** to support asynchronous embedding generation, retrieval benchmarking (BM25 + dense + reranking) and large-scale evaluation jobs across hybrid search systems.
- Refactored and productionized a behavioral inference pipeline by integrating **drift detection, bias analysis, ROC-based threshold calibration, and confidence driven intensity scaling**, reducing low-confidence personalization errors by **5%** and improving robustness under distribution shifts.

National University of Singapore (NUS)

Deep Learning Research Intern

Jul 2023 - Aug 2023

Singapore

- Collaborated with researchers to design and optimize a real-time image captioning system using a **ResNet–LSTM architecture**, training on **30,000+ image–caption pairs** with **GloVe 300D embeddings** and achieving **50+ captions/sec** inference throughput.
- Evaluated model performance on the **Flickr30k dataset**, achieving a **BLEU score of 0.62**; conducted **data preprocessing** and **model evaluation**.

VLOG Innovations

Artificial Intelligence Intern

Jan 2023 – Mar 2023

Chennai, India

- Trained and optimized a **YOLOv5 model** to detect PCB defects improving micro defect detection accuracy by **18%** over manual inspection and reducing false positives by **12%** through model tuning and workflow alignment.
- Led integration efforts by **partnering with QA and R&D teams** to deploy the model into PCB fabrication workflows, enabling real-time defect inspection across **100+ PCB samples**.

Projects

End-to-End Boston 311 Dashboard & Chatbot - [Link](#) | *BigQuery, FastAPI, Airflow, Docker, Cloud Run*

Aug 2025 – Dec 2025

- Architected an **AI-powered conversational system** enabling natural-language access to Boston's 311 data by fine-tuning **sqlcoder-7b-2** with **QLoRA** and integrating a **FAISS-based RAG pipeline** for schema-aware SQL generation and low-latency inference.
- Engineered scalable data pipelines using **Airflow** and **BigQuery** to process **280K+ annual service requests** (1M+ historical records), and deployed the end-to-end system on **Docker** and **Cloud Run** to support production-grade workloads.

NL2SQL using Retrieval-Augmented T5 - [Link](#) | *PyTorch, Hugging Face, LangChain, T5, RAG, FAISS*

Jun 2025 – Aug 2025

- Optimized a **T5-base** model for natural language-to-SQL translation on the Spider dataset, achieving **54.96% execution accuracy** using AdamW, warmup scheduling, gradient accumulation, and mixed precision training.
- Assembled a **retrieval-augmented GenAI pipeline** with LangChain, LlamaIndex, SentenceTransformer embeddings and FAISS operationalized via **Streamlit** to improve schema grounding and reduce hallucinations.

Candidate Recommendation Engine — [Link](#) | *Transformers, NLP, scikit-learn, Explainable AI*

Apr 2025 – Jun 2025

- Implemented an AI-driven resume ranking system leveraging **BERT NER**, **Sentence-Transformers**, and hybrid similarity scoring to evaluate **50+ resumes per run** with interpretable match percentages.
- Delivered a **Streamlit-based candidate recommendation engine** with Top-N ranking, hybrid semantic scoring, and interpretable fit explanations streamlining recruiter evaluation workflows.

Technical Skills

Languages

Python, SQL, Java, C, R

AI & Machine Learning

Large Language Models (LLMs), Transformers, Retrieval-Augmented Generation (RAG), Langchain, Lang-Graph, Explainable AI, Deep Learning, Natural Language Processing, Computer Vision

Libraries & Frameworks

PyTorch, TensorFlow, Hugging Face Transformers, Sentence-Transformers, scikit-learn, NumPy, Pandas

Platforms & Tools

Docker, FastAPI, Flask, Google Cloud (Vertex AI, Cloud Run), Redis, AWS, BigQuery, Airflow, MLflow, Git, GitHub, Jupyter, VS Code

Research & Activities

2nd Place – Boston 311 AI System, Google Cambridge : Presented a production-grade AI system to industry engineers.

Dec 2025

Graduated top of class in B.Tech Artificial Intelligence and Data Science

May 2024

Cognitive Defense: Cyber Attack Prediction and Security Design in ML Models (IEEE) : Co-authored IEEE paper on hybrid ML-based anomaly detection, achieving **97% accuracy**.

Jan 2024

IoT-based Smart Home Automation for Energy Conservation (IEEE) : Applied ML-driven optimization reducing energy consumption by **150 kWh**.

Oct 2023